Final exam · Unit 5 · Sturm-Liouville eigenproblems · 2.5 points · 50 min · Standard exam level

P19 — Sturm–Liouville problem with a Dirichlet end and a Robin end: positivity, the transcendental eigenvalue equation, large eigenvalues, orthogonality and norms

A rod $0 \le x \le 1$ has its end $x = 0$ kept at zero temperature, while its end $x = 1$ exchanges heat with surroundings at zero temperature by Newton's law, with a dimensionless transfer coefficient $h > 0$. Separating variables in the heat equation leads to the eigenvalue problem

$$X''(x) + \lambda X(x) = 0, \quad 0 < x < 1, \qquad X(0) = 0, \qquad X'(1) + h\,X(1) = 0.$$

(a) Prove that the problem has no eigenvalue $\lambda \le 0$. For $\lambda = \mu^2 > 0$ ($\mu > 0$) find the eigenfunctions and show that μ must satisfy $\tan\mu = -\mu/h$.

(b) Show graphically that there are infinitely many eigenvalues $0 < \lambda_1 < \lambda_2 < \dots$, locate μ_n in an interval of length $\pi/2$, and obtain the asymptotic behaviour of μ_n and λ_n for large n. For $h = 1$, give λ_1 and λ_2 to four significant figures and compare with the asymptotic formula.

(c) Prove, by integrating by parts (Green's identity), that eigenfunctions of different eigenvalues are orthogonal on $[0,1]$. Show that $\displaystyle\int_0^1 \sin^2(\mu_n x)\,dx = \frac{\mu_n^2 + h^2 + h}{2(\mu_n^2 + h^2)}$ and write the formula for the coefficients of the expansion $f(x) = \sum_n c_n \sin(\mu_n x)$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns: the eigenvalue $\lambda \in \mathbb{R}$ and the eigenfunction $X(x) \not\equiv 0$ on $[0,1]$, of class C^2.
- Linear, homogeneous, second-order ODE with constant coefficients; homogeneous boundary conditions: Dirichlet at $x = 0$, Robin (third kind) at $x = 1$ with $h > 0$.
- Data: $h > 0$ (general); $h = 1$ in the numerical questions.

Phase 2. Canonical formulation

Sturm-Liouville form (Haberman Sec. 5.3): $\frac{d}{dx}\left(p\frac{dX}{dx}\right) + qX + \lambda\sigma X = 0$ with $p = 1, q = 0, \sigma = 1$, and boundary conditions $\beta_1 X(0) + \beta_2 X'(0) = 0$ with $(\beta_1, \beta_2) = (1, 0)$, $\beta_3 X(1) + \beta_4 X'(1) = 0$ with $(\beta_3, \beta_4) = (h, 1)$. It is a regular Sturm-Liouville problem ($p, \sigma > 0$ on $[0,1]$), so it has real eigenvalues, one eigenfunction per eigenvalue, and orthogonal eigenfunctions with weight $\sigma = 1$.

Part (a) · 0.8 points

No eigenvalues $\lambda \le 0$ and the eigenvalue equation

Step 1. Case $\lambda = -\mu^2 < 0, \mu > 0$

Characteristic equation $r^2 - \mu^2 = 0, r = \pm\mu$, so $X = A\cosh(\mu x) + B\sinh(\mu x)$. $X(0) = A = 0$. Then $X' = B\mu\cosh(\mu x)$ and

$$X'(1) + hX(1) = B\big(\mu\cosh\mu + h\sinh\mu\big) = 0.$$

For $\mu > 0$ and $h > 0$ both terms in the bracket are positive, so $B = 0$ and $X \equiv 0$: no negative eigenvalues.

Step 2. Case $\lambda = 0$

$X'' = 0$, so integrating twice $X = Ax + B$. $X(0) = B = 0$; $X'(1) + hX(1) = A + hA = (1+h)A = 0$ with $1 + h > 0$, so $A = 0$: $\lambda = 0$ is not an eigenvalue.

Step 3. Cross-check with the Rayleigh quotient

Multiplying $X'' = -\lambda X$ by X and integrating by parts, $\int_0^1 XX''dx = [XX']_0^1 - \int_0^1 (X')^2 dx$, so (Haberman Sec. 5.6)

$$\lambda = \frac{-[XX']_0^1 + \int_0^1 (X')^2 dx}{\int_0^1 X^2 dx} = \frac{h\,X(1)^2 + \int_0^1 (X')^2 dx}{\int_0^1 X^2 dx},$$

using $X(0) = 0$ and $X'(1) = -hX(1)$. The numerator is ≥ 0, and it is 0 only if $X' \equiv 0$ and $X(1) = 0$, i.e. $X \equiv 0$. Hence $\lambda > 0$, in agreement with Steps 1-2.

Step 4. Case $\lambda = \mu^2 > 0$

$r^2 + \mu^2 = 0, r = \pm i\mu$, so $X = A\cos(\mu x) + B\sin(\mu x)$. $X(0) = A = 0$, $X = B\sin(\mu x)$, $X' = B\mu\cos(\mu x)$:

$$X'(1) + hX(1) = B\big(\mu\cos\mu + h\sin\mu\big) = 0.$$

A nontrivial solution needs $B \neq 0$, so $\mu\cos\mu + h\sin\mu = 0$. If $\cos\mu = 0$ this would give $h\sin\mu = \pm h = 0$, impossible; so we may divide by $h\cos\mu$:

$$\tan\mu = -\frac{\mu}{h}, \qquad X_n(x) = \sin(\mu_n x), \qquad \lambda_n = \mu_n^2.$$

KEY RESULTS

- $\lambda < 0$: $\mu\cosh\mu + h\sinh\mu > 0$ forces $X \equiv 0$. $\lambda = 0$: $(1+h)A = 0$ forces $X \equiv 0$.
- $\lambda = \mu^2 > 0$: $X_n = \sin(\mu_n x)$ with $\mu\cos\mu + h\sin\mu = 0$, i.e. $\tan\mu = -\mu/h$.

Part (b) · 0.8 points

Infinitely many eigenvalues and their asymptotic behaviour

Step 1. Graphical argument

For $\mu > 0$ the line $y = -\mu/h$ is negative and decreasing. The function $\tan\mu$ has branches on $\big((n-\tfrac12)\pi, (n+\tfrac12)\pi\big)$, on each of which it increases continuously from $-\infty$ to $+\infty$; on the first piece $(0, \pi/2)$ it is positive, so there is no root there. On the branch $n \geq 1$, $\tan\mu < 0$ exactly on $\big((n-\tfrac12)\pi, n\pi\big)$, and the difference $\tan\mu + \mu/h$ goes from $-\infty$ to $n\pi/h > 0$ there, increasing strictly (its derivative is $\sec^2\mu + 1/h > 0$). By the intermediate value theorem it vanishes **exactly once**:

$$\Big(n - \frac12\Big)\pi < \mu_n < n\pi, \qquad n = 1, 2, 3, \ldots$$

There are infinitely many branches, hence infinitely many eigenvalues $\lambda_n = \mu_n^2$, increasing to $+\infty$ (Haberman Sec. 5.3, property 2). Negative roots $-\mu_n$ give the same eigenfunctions up to sign and are not new.

Step 2. Large n

Write $p_n = (n-\tfrac12)\pi$ and $\mu_n = p_n + \varepsilon_n$ with $0 < \varepsilon_n < \pi/2$. Since $\cos p_n = 0$ and $\sin p_n = \pm 1$, the addition formulas give $\tan(p_n + \varepsilon) = \dfrac{\sin(p_n+\varepsilon)}{\cos(p_n+\varepsilon)} = \dfrac{\sin p_n\cos\varepsilon}{-\sin p_n\sin\varepsilon} = -\cot\varepsilon$. The eigenvalue equation becomes

$$-\cot\varepsilon_n = -\frac{p_n + \varepsilon_n}{h} \iff \tan\varepsilon_n = \frac{h}{p_n + \varepsilon_n}.$$

The right-hand side tends to 0, so $\varepsilon_n \to 0$ and $\tan\varepsilon_n = \varepsilon_n + O(\varepsilon_n^3)$: $\varepsilon_n = \dfrac{h}{p_n} + O(p_n^{-3})$. Squaring,

$$\lambda_n = \mu_n^2 = p_n^2 + 2p_n\varepsilon_n + \varepsilon_n^2 = \left(n - \frac{1}{2}\right)^2\pi^2 + 2h + O(n^{-2}).$$

Geometrically: the line becomes steeper relative to the branches, and the roots crowd towards the left asymptotes $(n-\frac{1}{2})\pi$ (Haberman Sec. 5.9). The leading term $(n-\frac{1}{2})^2\pi^2$ is the eigenvalue of the problem with $X(0)=0$, $X'(1)=0$: for large n the eigenfunctions oscillate so fast that the Robin end looks like an insulated end.

Step 3. Numbers for $h = 1$

Solving $\mu\cos\mu + \sin\mu = 0$ numerically (Newton's method or bisection) inside $(\pi/2, \pi)$ and $(3\pi/2, 2\pi)$:

n	**interval**	μ_n	λ_n	**asymptotic** μ_n	**asymptotic** λ_n
1	(1.571, 3.142)	2.029	4.116	$\frac{\pi}{2} + \frac{2}{\pi} = 2.207$	$\frac{\pi^2}{4} + 2 = 4.467$
2	(4.712, 6.283)	4.913	24.14	$\frac{3\pi}{2} + \frac{2}{3\pi} = 4.925$	$\frac{9\pi^2}{4} + 2 = 24.21$

The error of the asymptotic formula drops from about 9% to 0.3% between $n=1$ and $n=2$ for λ_n.

KEY RESULTS

- One root on each branch: $(n-\frac{1}{2})\pi < \mu_n < n\pi$, $n = 1, 2, \ldots$
- $\mu_n = (n-\frac{1}{2})\pi + \dfrac{h}{(n-\frac{1}{2})\pi} + O(n^{-3})$, $\lambda_n = (n-\frac{1}{2})^2\pi^2 + 2h + O(n^{-2})$.
- $h = 1$: $\mu_1 = 2.029$, $\lambda_1 = 4.116$; $\mu_2 = 4.913$, $\lambda_2 = 24.14$ (asymptotic: $\mu_1 \approx 2.207$, $\mu_2 \approx 4.925$).

Part (c) · 0.9 points

Orthogonality and norms

Step 1. Green's identity

For C^2 functions u, v, the product rule gives $\dfrac{d}{dx}(uv' - vu') = u'v' + uv'' - v'u' - vu'' = uv'' - vu''$. Integrating over $[0,1]$ with Barrow's rule (Haberman Sec. 5.5, Green's formula with $p=1$):

$$\int_0^1 (uv'' - vu'')\,dx = \big[uv' - vu'\big]_0^1 = \big(u(1)v'(1) - v(1)u'(1)\big) - \big(u(0)v'(0) - v(0)u'(0)\big).$$

Step 2. The boundary terms vanish

Take $u = X_m, v = X_n$ with $m \neq n$. At $x = 0$: $X_m(0) = X_n(0) = 0$, so the bracket is 0. At $x = 1$: $X_m'(1) = -hX_m(1)$ and $X_n'(1) = -hX_n(1)$, so

$$X_m(1)X_n'(1) - X_n(1)X_m'(1) = X_m(1)\big(-hX_n(1)\big) - X_n(1)\big(-hX_m(1)\big) = 0.$$

Step 3. Orthogonality

With the ODE, $X_n'' = -\lambda_n X_n$ and $X_m'' = -\lambda_m X_m$:

$$\int_0^1 (X_m X_n'' - X_n X_m'')\,dx = (\lambda_m - \lambda_n)\int_0^1 X_m X_n\,dx = 0.$$

Since $\lambda_m \neq \lambda_n$ (the μ_n lie in disjoint intervals), $\displaystyle\int_0^1 \sin(\mu_m x)\sin(\mu_n x)\,dx = 0$. Note that the μ_n are not multiples of a common number, so this is not the elementary orthogonality of $\sin(n\pi x)$: it holds because of the boundary condition. (Direct check in the script: the integral $\frac{n\sin m\cos n - m\cos m\sin n}{m^2-n^2}$ vanishes after substituting $\cos\mu = -h\sin\mu/\mu$.)

Step 4. Norm

With $\sin^2 s = \frac{1}{2}(1-\cos 2s)$ and the substitution $s = 2\mu_n x$, $ds = 2\mu_n\,dx$ for the cosine:

$$\int_0^1 \sin^2(\mu_n x)\,dx = \left[\frac{x}{2} - \frac{\sin(2\mu_n x)}{4\mu_n}\right]_0^1 = \frac{1}{2} - \frac{\sin 2\mu_n}{4\mu_n} = \frac{1}{2} - \frac{\sin\mu_n\cos\mu_n}{2\mu_n}.$$

From the eigenvalue equation $\cos\mu_n = -\dfrac{h\sin\mu_n}{\mu_n}$, so $-\dfrac{\sin\mu_n\cos\mu_n}{2\mu_n} = \dfrac{h\sin^2\mu_n}{2\mu_n^2}$. Squaring the same equation, $\mu_n^2\cos^2\mu_n = h^2\sin^2\mu_n$, i.e. $\mu_n^2(1-\sin^2\mu_n) = h^2\sin^2\mu_n$, so $\sin^2\mu_n = \dfrac{\mu_n^2}{\mu_n^2+h^2}$. Therefore

$$\int_0^1 \sin^2(\mu_n x)\,dx = \frac{1}{2} + \frac{h}{2(\mu_n^2+h^2)} = \frac{\mu_n^2+h^2+h}{2(\mu_n^2+h^2)}.$$

For $h = 1, n = 1$: $\dfrac{4.1159+1+1}{2(4.1159+1)} = 0.5977$. The norm is slightly above $\frac{1}{2}$ and tends to $\frac{1}{2}$ as $n \to \infty$.

Step 5. Expansion coefficients

If $f(x) = \sum_{n\geq 1} c_n \sin(\mu_n x)$, multiply by $\sin(\mu_m x)$, integrate over $[0,1]$ and use Step 3 (only $n = m$ survives; Haberman Sec. 5.3):

$$c_m = \frac{\int_0^1 f(x)\sin(\mu_m x)\,dx}{\int_0^1 \sin^2(\mu_m x)\,dx} = \frac{2(\mu_m^2 + h^2)}{\mu_m^2 + h^2 + h}\int_0^1 f(x)\sin(\mu_m x)\,dx.$$

KEY RESULTS

- $\int_0^1 (X_m X_n'' - X_n X_m'')dx = \left[X_m X_n' - X_n X_m'\right]_0^1 = 0$, hence $(\lambda_m - \lambda_n)\int_0^1 X_m X_n\,dx = 0$.
- $\int_0^1 \sin^2(\mu_n x)dx = \frac{1}{2} - \frac{\sin 2\mu_n}{4\mu_n} = \frac{\mu_n^2 + h^2 + h}{2(\mu_n^2 + h^2)}$; for $h = 1, n = 1$: 0.5977.
- $c_n = \frac{2(\mu_n^2 + h^2)}{\mu_n^2 + h^2 + h}\int_0^1 f(x)\sin(\mu_n x)\,dx$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $h \to 0^+$ (insulated end): $\tan\mu = 0^-$ pushes $\mu_n \to (n - \frac{1}{2})\pi$, the eigenvalues of $X(0) = 0$, $X'(1) = 0$, and the norm $\to \frac{1}{2}$. $h \to \infty$ (end held at zero): the line becomes horizontal, $\tan\mu \to 0^-$ from the left of $n\pi$, so $\mu_n \to n\pi$, the Dirichlet eigenvalues $n^2\pi^2$, and the norm $\to \frac{1}{2}$. For every $h > 0$, λ_n lies between these two Dirichlet-Neumann bounds.
- **Checks:** $\sin(\mu_n x)$ satisfies the ODE and both boundary conditions (script); the numerical roots lie in the predicted intervals; $\int_0^1 \sin(\mu_1 x)\sin(\mu_2 x)dx = 0$ numerically to 20 digits; the norm formula agrees with direct quadrature. In the heat problem the mode n decays as $e^{-k\lambda_n t}$, so $\lambda_1 = 4.116$ (for $h = 1$) sets the slowest decay rate.
- **Common mistakes in the exam:** (1) dropping the case $\lambda < 0$ or $\lambda = 0$ without proof, or "proving" it with the wrong sign of h; (2) writing $\mu_n = n\pi$ or $(n - \frac{1}{2})\pi$, as for Dirichlet or Neumann ends; (3) placing the roots in $(n\pi, (n + \frac{1}{2})\pi)$, where $\tan\mu > 0$; (4) using $\int_0^1 \sin^2 = \frac{1}{2}$ for the norm, which is false here because $\sin 2\mu_n \neq 0$.